

**CAN THO UNIVERSITY
COLLEGE OF INFORMATION TECHNOLOGY AND
COMMUNICATION**



**THESIS
SOFTWARE ENGINEERING**

**WEB APPLICATION FOR DENTAL CLINIC
APPOINTMENT MANAGEMENT**

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Cần Thơ, 09/2025

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Supervisor

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LIST OF ABBREVIATIONS

No.	Term/ Abbreviation	Definition/ Explanation
1	SRS	Software Requirements Specification – A document that describes the nature of a project, software, or application, including functional and non-functional requirements.
2	MEVN	MongoDB, Express.js, React.js, Node.js – A collection of JavaScript-based technologies used to develop web applications.
3	API	Application Programming Interface – A set of rules and protocols that allows different software applications to communicate with each other.
4	RBAC	Role-Based Access Control – A method of restricting network access based on the roles of individual users within an enterprise (e.g., Admin, Dentist, Receptionist).
5	LLM	Large Language Model – An AI model capable of understanding and generating human language, used in this project via the Gemini API.
6	IDE	Integrated Development Environment – Software for building applications that combines common developer tools into a single GUI (e.g., Visual Studio Code).
7	UI / UX	User Interface / User Experience – The visual elements of the application and the overall experience of the user when interacting with the system.
8	SPA	Single-Page Application – A web application that interacts with the user by dynamically rewriting the current web page with new data from the web server, instead of loading entire new pages.
9	JSON	JavaScript Object Notation – A lightweight data-interchange format that is easy for humans to read and write and easy for machines to parse and generate.
10	BSON	Binary JSON – A binary-encoded serialization of JSON-like documents, used by MongoDB for data storage.
11	SCM	Source Code Management – Tools used to track modifications to a source code repository (e.g., Git &

		GitHub).
12	CDM	Conceptual Data Model – A structured view of the data required to support business processes, independent of any physical storage.
13	PDM	Physical Data Model – A representation of a data design which takes into account the facilities and constraints of a given database system.
14	DOM	Document Object Model – A programming interface for web documents that represents the page so that programs can change the document structure, style, and content.

ABSTRACT

Context: In the context of rapid digital transformation, applying information technology to healthcare management has become urgent to enhance service quality. While modern technology offers immense potential for optimization, a significant number of small and medium-sized dental clinics still operate using manual processes, paper-based logbooks, or fragmented software tools. This reliance on outdated administrative methods frequently results in operational inefficiencies, such as prone-to-error scheduling, difficulties in retrieving patient medical history, and communication bottlenecks between staff and patients. These limitations ultimately lead to scheduling conflicts, wasted resources, and a diminished patient experience due to unnecessary waiting times and a lack of accessible service information.

Objective: The objective of this project is to build and develop a "Web Application for Dental Clinic Appointment Management." The system focuses on solving the core problem of optimizing the appointment scheduling process, allowing patients to actively choose doctors and time slots. Simultaneously, it provides a centralized and intuitive management tool for clinic staff to minimize errors and improve operational efficiency.

Methodology: The system is developed as a web-based application, utilizing the React.js library to construct a user-friendly and highly interactive interface. The backend is built on the Node.js platform combined with the Express framework, employing the MongoDB NoSQL database to flexibly store complex data such as electronic medical records and schedules. Notably, the system integrates an AI Chatbot feature to support basic inquiries and automatically guide patients through the booking process.

Result: The result of this project is a complete system with role-based access control for four user groups: Patient, Receptionist, Dentist, and Administrator. The application successfully fulfills key functions, including convenient online appointment booking, electronic medical record management, and automatic appointment reminders. This serves as a significant foundation for dental clinics to modernize their workflows and enhance customer satisfaction.

INTRODUCTION

1. Problem Statement

The use of technology in healthcare (E-health) has become a critical need to improve service quality in the context of the Fourth Industrial Revolution's intense digital transformation. However, there is still a paradox in many small and medium-sized dental clinics: appointment management procedures still mostly rely on manual methods or disjointed instruments, despite patients' growing expectations for convenience. Serious software engineering issues including data conflicts, information bottlenecks, and a lack of real-time interactions result from the lack of a centralized management system. Additionally, the current solutions available on the market are frequently complex, expensive ERP systems that do not fully satisfy the unique requirements of appointment management in a flexible and user-experience-optimized manner.

In order to close this gap, it is imperative to develop a software solution that is both practically significant and scientifically rigorous. The selection of the topic "Web Application for Dental Clinic Appointment Management" aims not only to address the urgency of automating the scheduling process, thereby eliminating human errors, but also to leverage modern technologies (Node.js, React.js, AI Chatbot) to create a smart operational ecosystem. With the ultimate goal of giving clinics an efficient management tool and guaranteeing patients a convenient, proactive medical experience, this is the main driving force behind the research group's system development.

2. Background and Related Works

Article [1] focuses on a mobile solution designed to optimize management processes in specialized dental centers. The authors utilized the Scrum methodology to develop an application that directly addresses long waiting times and errors inherent in manual management. Experimental results demonstrated superior operational efficiency: appointment registration time decreased by 44.13%, patient attendance rates increased by 17.58%, and user satisfaction rose by 65%. While successful in enhancing user experience, the study suggests necessary future developments, such as integrating electronic medical records, automated payments, and telemedicine systems, to create a more comprehensive healthcare ecosystem.

Article [2] approaches the problem through an integrated web platform that combines online booking functionality with an e-commerce module for selling dental care products. This system was developed using the Waterfall model and evaluated based on ISO 25010 international software quality standards, featuring the application of predictive analytics to forecast future demand and revenue. However, the research also highlights certain technical limitations of the current system, including support for

only a single-branch setup, the lack of an integrated online payment gateway, and the inability to function offline without network connectivity .

3. Research Objectives

General Objective:

The primary objective of this project is to fully develop the WebApp to assist dental clinics in transitioning from manual management methods (such as logbooks or Excel) to a modern software solution. The ultimate goal is to create a tool that makes it easier for patients to proactively book appointments, while simultaneously relieving clinic staff from the burden of scheduling, thereby minimizing errors such as double-booking or missed appointments.

Specific Objectives:

- Develop Appointment Booking and Management Features
- Digitalize Medical Records
- Integrate Automated Support (AI & Notifications)
- Provide Management Tools for Clinic Owners

4. Research Subjects and Scope

4.1. Research Subjects

Based on the objectives and scope of the project, the primary research subjects include:

- **The DCAM-WebApp Software System:** Focusing on researching the architecture and development of the web-based dental clinic operational application, with a specific emphasis on the online appointment booking subsystem and electronic medical record management.
- **Clinic Business Workflows:** Analyzing and modeling actual operational workflows, including the customer booking process, the staff approval/scheduling process (to avoid time conflicts), and the medical history storage process .
- **System User Groups:** Researching the characteristics, specific needs, and access permissions for the four main groups directly interacting with the system: Patients, Receptionists, Dentists, and Administrators.
- **Technology and Integration Techniques:** Researching the application of the MEVN stack (MongoDB, Express, React.js, Node.js) and the integration of an AI Chatbot to solve problems related to real-time interaction and automated customer support.

4.2. Research Scope

Functional Scope:

- The system focuses on developing core functionalities for a single dental clinic, including: Appointment Management (primary focus), Electronic Medical Record Management, Staff Management, and Basic Statistical Reporting.
- Out of scope: The project does not cover multi-clinic chain management, in-depth health insurance processing, or complex material supply chain management. Features such as online payment and SMS notifications will be implemented in a test/sandbox environment solely to simulate the workflow.

Technical Scope:

- The application is built on a Web-based platform, ensuring compatibility with common web browsers on both desktop and mobile devices.
- The project does not involve developing separate Native Apps for iOS or Android at this stage.

Geographical and Operational Scope:

- The system is specifically designed for the Vietnamese market, utilizing Vietnamese as the default display language and adhering to local standards for date/time (DD/MM/YYYY) and currency (VND) formatting .
- The target users are small to medium-sized dental clinics where current management processes are still largely manual or fragmented.

5. Research Methods and Content

5.1. Research Methods

During the implementation of this term project, the study was conducted based on three main research methods: literature review, observation and system analysis, and experimentation.

Theoretical Review Method

The group studied theories about software architecture, core technologies of MEVN Stack (MongoDB, Express.js, React.js, Node.js) to establish the technical foundation of the application and SRS standards and regulations for medical data security to ensure that the system meets the technical and operational standards of a medical application.

Observation and System Analysis Method

This method was employed to observe the business and technical operations of the dental clinics and to elicit functional and non-functional requirements. The business operations were then analyzed to model the processes of appointment booking and medical record management through the use of Use Case diagrams and specification analysis tables. At the same time, the team analyzed the four types of users (Patient/ Receptionist/ Dentist/ Administrator) in order to design suitable interfaces.

Experimentation Method

This method was employed in the database design, programming, and testing of the DCAM-WebApp. After the DCAM-WebApp was built and integrated with the AI Chatbot and notification modules, it was deployed on a local server to test its performance in terms of transactions and stability before release.

5.2. Research Content

5.2.1. Implementation Process

The development process of the WebApp follows the Waterfall Model [3]. This is a linear software development model where phases are executed sequentially, and the output of one phase serves as the input for the next. The process is divided into four main phases as follows:

Phase 1: Requirements Analysis

- Objectives: To understand the actual operational workflows at dental clinics and identify the needs of relevant stakeholders.
- Activities:
 - Gather information regarding appointment booking, medical record management, and staff management processes.
 - Analyze the SRS to define Functional and Non-functional requirements.
 - Identify four main user actors: Patient, Receptionist, Dentist, and Administrator.
- Expected Outcomes: A complete SRS document, including a detailed list of Use Cases and workflow diagrams.

Phase 2: System Design

- Objectives: To transform business requirements into a possible technical design.
- Activities:
 - Database Design: Construct the Logical Data Model and design the MongoDB Schema to store patient information, appointments, and invoices.
 - Architecture Design: Define the Client-Server architecture, utilizing RESTful APIs for communication between the Frontend (React.js) and Backend (Node.js).
 - Interface Design: Draft user interfaces to ensure user-friendliness and usability on both desktop and mobile devices.
- Expected Outcomes: System architecture diagrams, Database Schema, and Interface Mockups.

Phase 3: Implementation

- Objectives: To build the software source code based on the design specifications.
- Activities:
 - Backend Development: Develop APIs to handle business logic, integrate the AI Chatbot module, and manage user authentication.
 - Frontend Development: Program the website interface, incorporating features for booking, record viewing, and the administration dashboard.
 - Integration: Connect the Frontend with the Backend and integrate third-party services.
- Expected Outcomes: Complete source code and a functional application executable in a local server environment (Localhost).

Phase 4: Testing and Evaluation

- Objectives: To ensure the system operates stably, functions correctly, and is free of critical errors.
- Activities:
 - Perform Unit Testing for API functions.
 - Conduct Functional Testing focusing on critical flows: Appointment Booking, Conflict Handling, and Automated Chatbot Responses.
 - Evaluate performance and user experience.
- Expected Outcomes: Testing reports and a packaged software version ready for reporting.

5.2.2. Technologies for Application Development

To ensure the system is robust, scalable, and capable of real-time interaction, the following technologies and tools have been selected for the development of the project:

Frontend Technologies:

- React.js: A progressive JavaScript framework used for building the user interface. It is chosen for its component-based architecture and ease of integration.
- Axios: A promise-based HTTP client used to handle requests between the Frontend and the Backend APIs.

Backend Technologies:

- Node.js: A JavaScript runtime environment that executes JavaScript code outside a web browser, chosen for its non-blocking I/O model suitable for real-time applications.
- Express.js: A minimal and flexible Node.js web application framework used to build robust APIs and handle server-side routing.

- Gemini API: An external service integrated to power the AI Chatbot for automated patient support.

Database Management:

MongoDB: A NoSQL database used for its flexibility in handling unstructured data such as medical records and conversation logs.

Testing Tools:

Postman: A comprehensive API testing tool used to verify the functionality of RESTful endpoints during the backend development phase.

Version Control and Deployment Tools:

- Git & GitHub: Used for source code management (SCM) and version control to facilitate collaboration and track changes throughout the project.
- Visual Studio Code: The primary IDE used for coding.

6. Key Contributions of the Project

The research project delivers significant contributions to both the practical operation of dental clinics and the field of software engineering by providing a comprehensive solution known as the Dental Clinic Appointment Management WebApp. This system replaces fragmented manual management methods with a unified platform that tightly integrates appointment data, medical records, and staff information, thereby minimizing data redundancy and human errors. A key technical advancement is the implementation of automated conflict resolution logic within the scheduling process, which, along with automated notifications, significantly optimizes dentist workflows and reduces unannounced patient absences. Additionally, the project distinguishes itself through the integration of an AI Chatbot, an innovation that overcomes the time constraints of human staff by providing 24/7 real-time consultation and booking support, marking a substantial improvement over traditional legacy systems.

7. Thesis Structure

In addition to the standard sections such as the Table of Contents, List of Figures, List of Tables, and List of Abbreviations, the report is organized into the following main parts:

Part I: Introduction: Presents the problem statement and reviews existing solutions. Based on this analysis, the project objectives are defined, and a plan is established to develop the web application to resolve the identified problem. Furthermore, this section identifies the research subjects, the project scope, and the technologies utilized.

Part II: Content: Provides detailed information regarding the problem description, system design, solution implementation, and testing.

Chapter 1: Problem Description of Developing a Dental Clinic Appointment Web Application - Describes the current situation in dental clinics, outlines the problems caused by the use of manual scheduling and distributed management systems,

such as overlapping of appointments or duplication of information, and establishes the functional and non-functional requirements that any new system would address in response to the problems identified above.

Chapter 2: Theoretical Foundation - Presents an overview of the technologies, tools, and algorithms employed in the development of the project. The content of this chapter provides the necessary theoretical foundation for the analysis, design, and development of the system.

Chapter 3: Solution Design and Implementation - Details the system architecture, database schema, and user interface design. It describes the step-by-step implementation of core modules, including the online booking system with conflict resolution logic, electronic medical records management, and the integration of the AI Chatbot.

Chapter 4: Testing and Evaluation - Outlines the testing strategies applied, including unit and functional testing, to verify system stability and reliability. It presents the results of critical test cases—specifically regarding appointment scheduling and real-time interactions—and concludes with an evaluation of the system's performance against the initial project objectives.

Part III: Conclusion: Presents the results achieved upon the completion of the project, outlining the system's strengths as well as its remaining limitations, and thereby proposing directions for future development.

References: The References section lists the sources and authors of the materials consulted for the construction and development of the system.

MAIN CONTENT

CHƯƠNG 1. PROBLEM DESCRIPTION OF DEVELOPING A DENTAL CLINIC APPOINTMENT WEB APPLICATION

1. Introduction

The product developed within the scope of this research is the "Dental Clinic Appointment Management Web Application" (DCAM-WebApp). This is a newly developed independent software system, designed with the objective of completely replacing current cumbersome manual management processes, such as paper-based records or fragmented Excel spreadsheets. DCAM-WebApp does not function as a supplementary module but rather operates as a centralized "All-in-one" management platform, seamlessly integrating core functions ranging from online appointment scheduling and electronic medical record management to personnel coordination. The system operates independently on its own database while maintaining scalability by connecting with external services, such as the AI Chatbot and Email services, through standardized API.

The system is designed to facilitate interaction for four distinct user groups, each with specific characteristics and permissions. Patients serve as end-users with infrequent usage frequency, requiring an intuitive and simple interface to perform quick booking operations without necessitating advanced technical knowledge. Conversely, Receptionists and Dentists act as internal users with high and continuous usage frequency; while Receptionists require immediate system response to handle administrative workflows and payments, Dentists demand absolute accuracy and privacy when accessing medical records. Finally, Administrators hold the highest level of access rights to manage system configuration, user accounts, and statistical reporting, requiring a deeper understanding of the software's operational processes.

Regarding the technical environment, DCAM-WebApp operates based on a Client-Server architecture, and within the experimental scope of this project, the system is deployed in a local-hosted server environment. The server requires minimum hardware specifications of a 4-core CPU and 8GB RAM running on the Windows 11 operating system, along with a software platform that includes the Node.js Runtime and MongoDB database management system. On the client side, users can access the application via modern web browsers (Google Chrome, Microsoft Edge, Firefox) across cross-platform devices ranging from desktops to mobile phones. Communication between components is executed via secure HTTP/HTTPS protocols, and a stable Internet connection is mandatory to ensure the accurate functioning of AI integration and real-time notification features.